In [21]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

In [22]:
```
df = pd.read_csv("IRIS.csv")
```

In [23]:
```
df.head()
```

Out[23]:

	sepal_length	sepal_width	petal_length	petal_width	species
0	5.1	3.5	1.4	0.2	Iris-setosa
1	4.9	3.0	1.4	0.2	Iris-setosa
2	4.7	3.2	1.3	0.2	Iris-setosa
3	4.6	3.1	1.5	0.2	Iris-setosa
4	5.0	3.6	1.4	0.2	Iris-setosa

In [24]:
```
df.mean()
```

Out[24]:
```
sepal_length    5.843333
sepal_width     3.054000
petal_length    3.758667
petal_width     1.198667
dtype: float64
```

In [25]:
```
df.median()
```

Out[25]:
```
sepal_length    5.80
sepal_width     3.00
petal_length    4.35
petal_width     1.30
dtype: float64
```

In [26]:
```
df.mode()
```

Out[26]:

	sepal_length	sepal_width	petal_length	petal_width	species
0	5.0	3.0	1.5	0.2	Iris-setosa
1	NaN	NaN	NaN	NaN	Iris-versicolor
2	NaN	NaN	NaN	NaN	Iris-virginica

In [27]:
```
df.std()
```

Out[27]:
```
sepal_length    0.828066
sepal_width     0.433594
petal_length    1.764420
petal_width     0.763161
dtype: float64
```

In [28]:
```
df.min()
```

Out[28]:
```
sepal_length            4.3
sepal_width               2
petal_length              1
petal_width             0.1
species         Iris-setosa
dtype: object
```

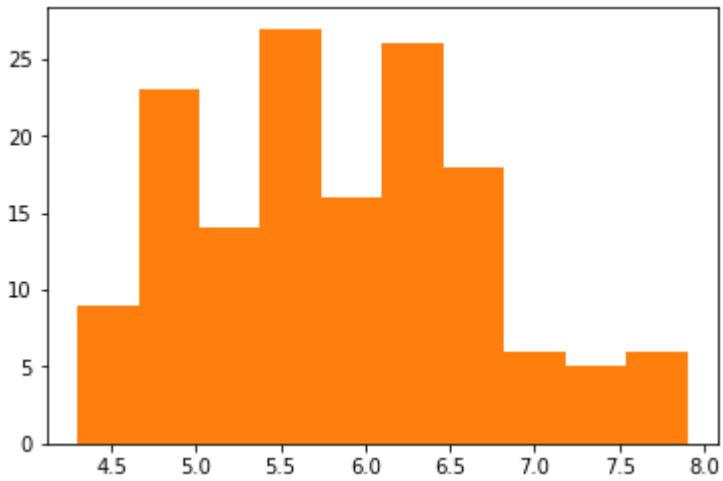
In [29]:
```
df.max()
```

Out[29]:
```
sepal_length               7.9
sepal_width                4.4
petal_length               6.9
petal_width                2.5
species         Iris-virginica
dtype: object
```

In [30]:
```
df.var()
```

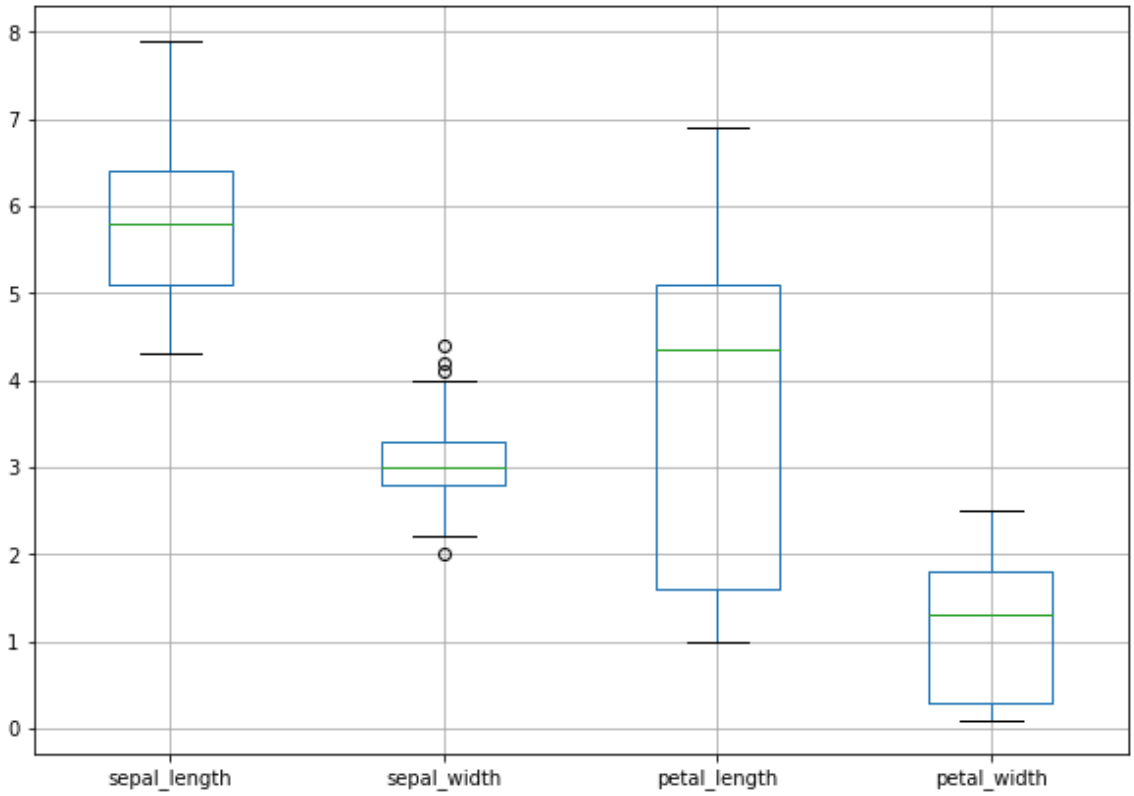
Out[30]:
```
sepal_length    0.685694
sepal_width     0.188004
petal_length    3.113179
petal_width     0.582414
dtype: float64
```

In [35]:
```
plt.hist(df['sepal_length'])
plt.show()
```



In [43]:
```
plt.figure(figsize = (10,7))
df.boxplot()
plt.show()
```

```
<matplotlib.figure.Figure at 0x7f68b933f750>
```



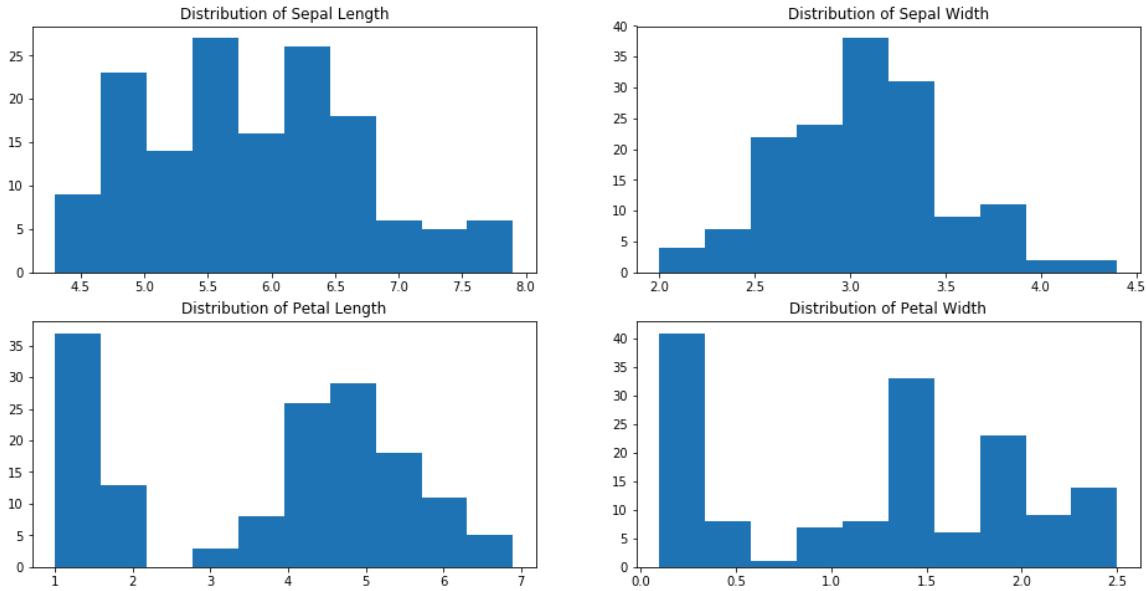
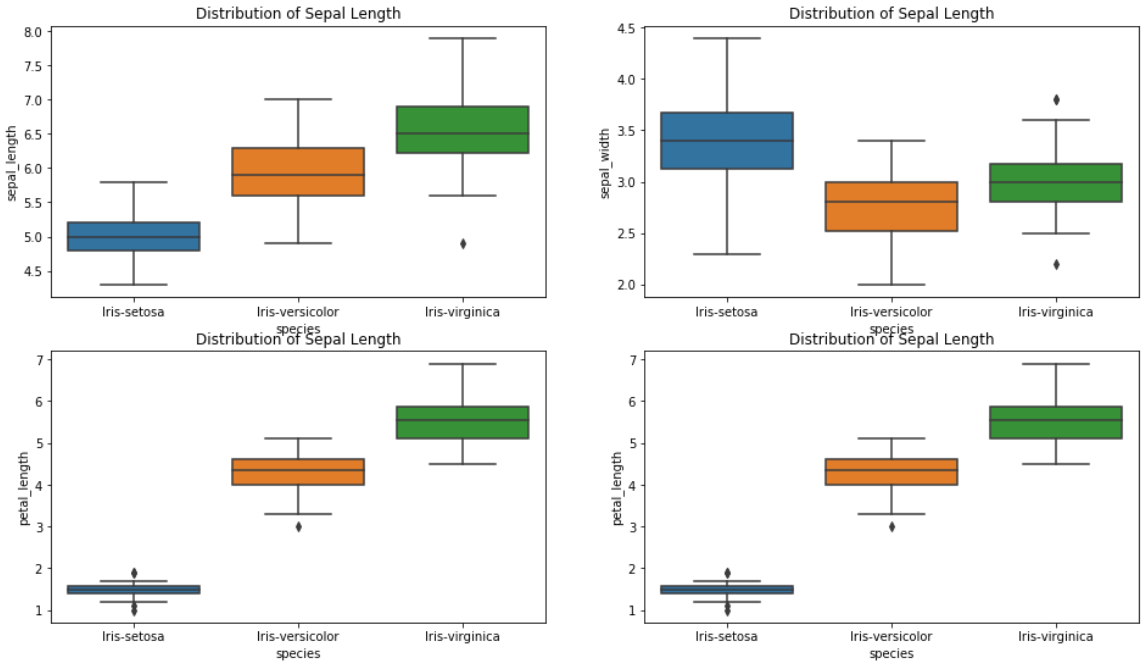
In [48]:
```
fig, axes = plt.subplots(2, 2, figsize=(16,8))

axes[0,0].set_title("Distribution of Sepal Length")
axes[0,0].hist(df["sepal_length"]);

axes[0,1].set_title("Distribution of Sepal Width")
axes[0,1].hist(df["sepal_width"]);

axes[1,0].set_title("Distribution of Petal Length")
axes[1,0].hist(df["petal_length"]);

axes[1,1].set_title("Distribution of Petal Width")
axes[1,1].hist(df["petal_width"]);
plt.show()
```



In [50]:
```
fig, axes = plt.subplots(2, 2, figsize=(16,9))
axes[0,0].set_title("Distribution of Sepal Length")
sns.boxplot(y="sepal_length", x="species", data=df, orient='v', ax=axes[0,
0])
axes[0,1].set_title("Distribution of Sepal Length")
sns.boxplot(y="sepal_width", x="species", data=df, orient='v', ax=axes[0,
1])
axes[1,0].set_title("Distribution of Sepal Length")
sns.boxplot(y="petal_length", x="species", data=df, orient='v', ax=axes[1,
0])
axes[1,1].set_title("Distribution of Sepal Length")
sns.boxplot(y="petal_length", x="species", data=df, orient='v', ax=axes[1,
1])
plt.show()
```